

Carbon Dioxide Conversion to Methanol over Size-selected Cu₄ Clusters at Low Pressures

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Supporting Information Placeholder

ABSTRACT: The activation of CO₂ and its hydrogenation to methanol is of much interest as a way to utilize captured CO₂. In this work we investigate the use of size-selected Cu₄ clusters supported on Al₂O₃ thin films for CO₂ reduction in the presence of hydrogen. The catalytic activity was measured under near-atmospheric reaction conditions with a low CO₂ partial pressure, and the oxidation state of the clusters was investigated by *in situ* grazing incidence X-ray absorption spectroscopy. The results indicate that size-selected Cu₄ clusters are the most active low-pressure catalyst for catalytic CO₂ conversion to CH₃OH. Density functional theory calculations reveal that Cu₄ clusters have a low activation barrier for conversion of CO₂ to CH₃OH. This study suggests that small Cu clusters may be excellent and efficient catalysts for recycling of released CO₂.

The industrial process of methanol (CH₃OH) synthesis from syngas (CO, CO₂ and H₂) is carried out at high pressures (10 to 100 bar) using a Cu/ZnO/Al₂O₃ catalyst¹. Due to increasing emission of CO₂ from fossil fuel combustion and other anthropogenic activities, this catalytic system has also become the focus of interest for obtaining sustainable CH₃OH by hydrogenation of captured CO₂ (CO₂ + 3H₂ → CH₃OH + H₂O)^{2,3}. Efforts have been made to modify and improve the industrial Cu/ZnO/Al₂O₃ catalyst³⁻⁵. Nevertheless, the high pressure required for achieving a quality yield of CH₃OH using these catalysts brings a great challenge for reducing the energy input and cost for this process. Also, effective catalysts are in need for alternative feed streams with lower CO₂ concentrations. Thus, developing an effective low-pressure catalyst for CO₂ reduction to CH₃OH is highly attractive.

Recently, size-selected subnanometer transition metal clusters have received considerable attention in catalysis,

because of their unique electronic and catalytic properties, which differ from bulk metal surfaces and larger nanoparticles. Although a number of computational and experimental studies have been focused on catalytic and electrocatalytic CO₂ reduction to fuels on various metal clusters⁶⁻⁸ and larger nanoparticles⁹⁻¹³, there is a paucity of research on CH₃OH synthesis from CO₂ and H₂ on size-selected non-precious metal clusters. Previously, we have successfully synthesized subnanometer metal clusters with narrow size distributions on thin film support materials (e.g., Al₂O₃ and Fe₃O₄), and these materials have shown great potential in the catalytic conversion of small molecules.¹⁴⁻¹⁶ In this contribution, we report on Al₂O₃ supported Cu₄ clusters as an effective catalyst for the reduction of CO₂ to CH₃OH at a low CO₂ partial pressure (0.013 atm), with a higher activity than those of recently developed low-pressure catalysts.

Cu₄ clusters were synthesized using size-selected cluster source,¹⁷ which enables single-size mass selection with atomic precision without fragmentation.¹⁸ The Cu₄ cluster was chosen based on preliminary density functional (DFT) calculations indicating they are active for methanol formation. By soft landing, 7 ng of Cu₄⁺ clusters were deposited onto a three monolayer amorphous Al₂O₃ thin film prepared by atomic layer deposition (ALD) on top of the native oxide of silicon wafer (SiO₂/Si(100)). The detailed description of the preparation method can be found in the previous report.¹⁷ Previous studies have also shown that such a film can keep a variety of clusters from sintering under reaction conditions.^{15,16,19} All samples were exposed to air after synthesis and oxidized copper clusters were identified in the subsequent characterization. The catalytic testing and grazing-incidence X-ray absorption near edge structure (GIXANES) measurements of Al₂O₃ supported Cu₄ clusters were performed in a home-built reaction cell^{14,20,21} which allows for X-ray scattering off the sample surface at a grazing incident angle ($\alpha_c = 0.18^\circ$) (Details see SI). All measurements were carried out

under *in-situ* conditions with 20 sccm flow of 1% CO₂ and 3% H₂ gas mixture carried in helium at a total pressure of 1.25 atm.

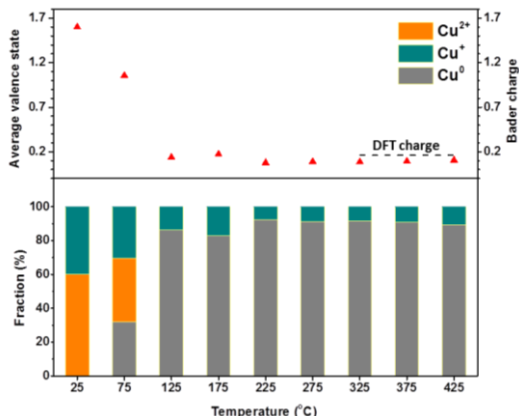


Figure 1. Oxidation state of Al₂O₃ supported Cu₄ clusters at different temperatures under *in-situ* catalytic reaction conditions (3% H₂, 1% CO₂ and 96% He, 1.25 atm): a) Average valence state of Cu₄ clusters. The inserted dashed line indicates the calculated charge from DFT for comparison. b) Linear combination fit (LCF) results of XANES spectra using bulk Cu foil, Cu₂O and CuO powder as reference materials.

The characteristic GIXANES features at the Cu K-edge²² reflect the evolution of the oxidation state of supported Cu₄ clusters under *in-situ* reaction conditions, as displayed in Figure S1(a). By comparing the absorption edge of the reference spectra (Cu foil, Cu₂O and CuO bulk standards in Figure S1(b)), a gradual reduction of Cu₄ clusters under reaction conditions (CO₂/H₂ gas feed) can be identified upon heating to 425 °C. To show the trend in the evolution of the average valence state of copper clusters, a quantitative linear combination fit (LCF) is performed using Cu (Cu⁰), Cu₂O (Cu⁺) and CuO (Cu²⁺) bulk standards. Figure 1 presents the LCF results as well as the average valence state of Cu₄ clusters with increasing reaction temperature. At room temperature, Cu₄ clusters can be described as primarily composed of oxidized Cu²⁺ (~60%) and Cu⁺ (~40%), with an average valence state of 1.60. At 75 °C a partial reduction takes place, yielding an average copper valence of 1.05. Starting at 125 °C, the Cu₄ clusters are practically fully reduced as indicated by the ~0.1 average valence state (Figure 1). The observed non-zero value can be ascribed to some charge transfer to a fully reduced (i.e., Cu⁰) cluster from either the alumina support or adjacent acidic hydroxyls on the surface.^{23,24}

The turn-over rate (TOR) of methanol formation from CO₂ hydrogenation over supported Cu₄ clusters is shown in Figure 2. Here, the TOR is defined as the yield of methanol formed per copper atom per second. As discussed earlier, under the reaction conditions, the catalyst becomes Cu⁰ when 125 °C is reached. This is also the temperature at which CH₃OH starts to be produced, suggesting that the reduction of CO₂ to CH₃OH is mainly catalyzed by the fully reduced state of the Cu₄ clusters. As displayed in Figure 2, the maximal TOR of ~4 × 10⁻⁴ molecule · s⁻¹ · atom⁻¹ was obtained at 225 °C and the rate of methanol production drops above 325 °C, which falls into the thermodynamic-control regime.²⁵ The

observed TOR for methanol synthesis is fairly high in comparison to the numbers in the literature.^{25,26} Table 1 lists the TOR's of CO₂ reduction to CH₃OH for the present catalyst (Al₂O₃ supported Cu₄ clusters), recently developed low-pressure catalyst (Ni₅Ga₃/SiO₂), as well as bulk Cu materials (Cu/ZnO/Al₂O₃ and polycrystalline Cu foil). It is clear that all the materials reached the maximum activity of CH₃OH in a similar temperature range, 200–240 °C. Compared to Ni₅Ga₃/SiO₂ and Cu/ZnO/Al₂O₃ at the ambient pressure, our catalyst demonstrated much higher activity (one order of magnitude higher) at a slightly higher total pressure (1.25 atm), but with much lower partial pressures (down by up to two orders of magnitude) of the reaction gases (H₂ and CO₂), although this activity of the supported Cu₄ clusters is lower than that of the polycrystalline Cu surface under a 6 times higher pressure (Table 1). It clearly shows an outstanding activity of the supported Cu₄ clusters for CO₂ reduction to CH₃OH at a low pressure.

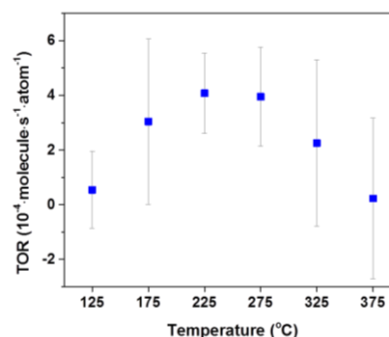


Figure 2. Turn-over rate (TOR) of CO₂ reduction to CH₃OH over Al₂O₃ supported Cu₄ clusters. The data plot is converted from Fig. S2(b) (See SI), by averaging 100 data points at each temperature.

In terms of methane formation (CO₂ + 4H₂ → CH₄ + 2H₂O), no CH₄ (m/z 15) was obtained as a product below 325 °C, see Figure S2. This indicates that methanation is not favorable in this temperature range. Nevertheless, we obtain an increasing CH₄ signal at 375 °C accompanied by a rise of m/z 18 signal of water, which likely implies the preference of methanation over methanol synthesis at higher temperatures. However, above 375 °C, we also observed the background desorption of hydrocarbon traces (m/z 43 and 56). We note that the fragmentation pattern of these hydrocarbons may also contribute to the CH₄ (m/z 15) signal in the mass spectrometer. Thus, to draw an affirmative conclusion on the high-temperature selectivity of CH₄, further experiments, e.g., with higher sample loading/purified gas feed, are needed, which is beyond the scope of this paper.

Copper may also catalyze a reverse water-gas shift reaction (rWGS, CO₂ + H₂ → CO + H₂O)^{26,27} at high temperatures. A clear assignment of CO (m/z =28) with mass spectrometry is not feasible in our experiments due to its overlap with fragment ions of CO₂ in the feedstock. However, we observe an increasing water signal m/z 18) above 375 °C when the TOR of methanol synthesis declines (see SI). This is a strong indication of other reaction pathways for CO₂ conversion at elevated temperatures, e.g., rWGS or methanation.

DFT calculations were carried out to help understand the

Table 1. Comparison of turn-over rates (TOR)/turn-over frequencies (TOF) of the present work and previous studies

Catalyst	Temperature	Total Pressure	Partial Pressure of H ₂	Partial Pressure of CO ₂	Maximum TOR/TOF of CH ₃ OH (molecule·s ⁻¹ ·atom ⁻¹)	Reference
Cu ₄ /Al ₂ O ₃	225°C	1.25 atm	0.038 atm	0.013 atm	4.0×10 ⁻⁴ (TOR)	This work
Ni ₅ Ga ₃ /SiO ₂	200 - 220°C	1 atm	0.75 atm	0.25 atm	6.7×10 ⁻⁵ (TOF)*	²⁵
Cu/ZnO/Al ₂ O ₃	200 - 220°C	1 atm	0.75 atm	0.25 atm	6.7×10 ⁻⁵ (TOF)*	²⁵
Polycrystalline Cu foil	237°C	5 atm	4.6 atm	0.4 atm	1.2×10 ⁻³ (TOR)	²⁶

*These values were converted from the graphs in ref ²⁵.

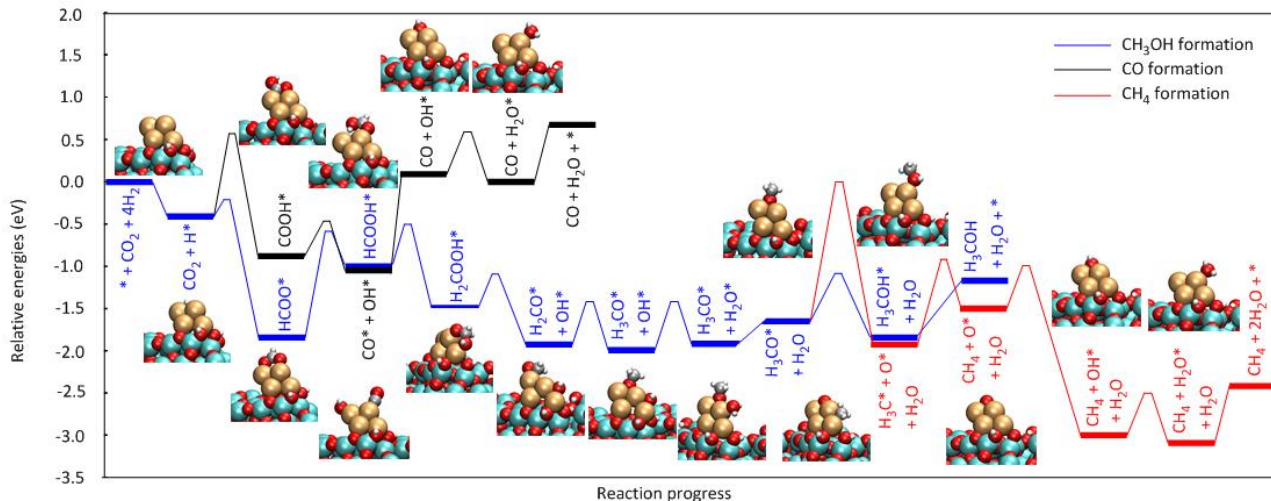


Figure 3. Calculated reaction pathways of CO₂ reduction to CH₃OH, CO and CH₄ on Al₂O₃ supported Cu₄ clusters. The catalyst surface site is labeled as “*”. To improve legibility, “H₂” was omitted from the labels after the initial state.

reaction mechanism of the catalytic reduction of CO₂. The Al₂O₃ supported Cu₄ cluster (Cu₄/Al₂O₃) was constructed by binding the Cu₄ cluster onto a hydroxylated amorphous Al₂O₃ surface model obtained from a molecular dynamics (MD) simulation²⁸. A GGA_PBE functional²⁹ with a plane wave basis set was used for geometry optimization and energy calculations. (see SI). The average charge on the supported Cu₄ cluster is found to be +0.16 |e|/atom using Bader charge analysis³⁰. In the cluster, the two Cu atoms that are bound to the surface oxygens/hydroxyls have slightly positive charges while the other two Cu atoms carry no charge (see SI). This agrees well with the experimental measurements that the catalyst is mainly fully reduced Cu (Figure 1). The slightly positive charges on the bound Cu atoms are due to the partial oxidation of Cu by the bridging O atoms and the hydroxyl groups on the support. The calculated reaction pathways for the CO₂ reduction to CH₃OH, CO and CH₄ on Cu₄ are shown in Figure 3. In general, the calculated reaction mechanism of CH₃OH formation on a supported Cu₄ cluster is similar to that on bulk Cu materials as reported in previous investigations^{3,8,31}. The initial step of CH₃OH formation is the formation of the HCOO* species (“*” represents an adsorbed species). The hydrogenation of the HCOO* species produces HCOOH*, which is further hydrogenated to H₂COOH*. Cleavage of the C-OH bond of H₂COOH* leads to its dissociation into H₂CO* and OH*, followed by further hydrogenation of H₂CO* to H₃CO* and OH* to H₂O*. The hydrogenation of H₃CO* generates the final product, CH₃OH. The rate-limiting step of the CH₃OH pathway on Cu₄ is found to be the hydrogenation of the HCOO* species with a predicted

barrier of 1.18 eV. This barrier for the Cu₄ cluster is lower than the predicted rate-limiting barriers for the Cu (111) surface (1.60 eV) and the Cu₂₀ cluster (1.41 eV), both of which were calculated using a GGA_PW functional by Yang *et al.*⁸ It is also notable that the reaction pathway of the Cu₄ cluster is energetically lower-lying than that of the Cu₂₀ cluster, which is lower than that of the Cu(111) surface.⁸ This indicates that the Cu₄ cluster could be more active for CH₃OH formation compared to bulk Cu surfaces and larger Cu nanoparticles.

The low reaction barrier for Cu₄ can be explained by the adsorption strength of the adsorbate species to the catalyst. Under-coordinated Cu sites in Cu₄ clusters lead to strong adsorption energies of the adsorbates as found in previous studies of subnanometer clusters,¹⁴ resulting in an energetically lower lying reaction pathway and a lower barrier. Previous work³⁻⁵ has illustrated that high CO₂ and H₂ pressure increase adsorption energies and make the formation of methanol energetically more favorable. This is why high pressures are needed for CO₂ reduction on Cu surfaces.³⁻⁵ It is further consistent with how the strong adsorption of the adsorbates on the under-coordinated Cu₄ clusters leads to high activity at a low CO₂ and H₂ pressures.

For comparison, the reaction pathways for the CO₂ reduction to CO (rWGS) and CH₄ were also investigated (Figure 3). In agreement with a previous DFT study,³² our calculations showed that the formation of COOH* on Cu₄, which initiates rWGS, has much higher barrier (1.08 eV) than that of HCOO* (0.18 eV), and the rWGS pathway is energetically higher-lying than the CH₃OH pathway. This suggests that the CH₃OH formation is likely to be predominant at lower

temperatures for Cu₄. However, at higher temperatures rWGS could become significant due to a bigger reaction rate consistent with the decreasing signal of CH₃OH above 325 °C (Figure 2). CH₄ formation, on the other hand, follows the same path as CH₃OH formation to form the H₃CO* species. Then, the breaking of the C-O bond of H₃CO* leads to H₃C* and O*, after which H₃C* is hydrogenated to CH₄ and O* is finally hydrogenated to H₂O. The rate-limiting step of the CH₄ pathway (H₃CO* → CH₃* + O*) has a much higher barrier (1.69 eV) than that of the CH₃OH pathway, suggesting that the CH₄ formation would require a much higher reaction temperature than CH₃OH formation. This result supports our experimental observations that CH₃OH was obtained as the main product at a lower temperature range (<375°C), and explains why Cu clusters favor the CH₃OH formation by the gas-phase hydrogenation of CO₂ rather than the CH₄ formation, even though the net reaction of the latter is thermodynamically more favorable.

To conclude, to our best knowledge the Al₂O₃ supported size-selected Cu₄ clusters exhibit the highest reported activity to date for CO₂ reduction to CH₃OH at a low CO₂ partial pressure. The unique coordination environment of Cu atoms in size-selected subnanometer clusters results in the active sites that are superior to those of larger Cu particles. These results for size-selected Cu clusters demonstrate their great potential for the development of novel low-pressure catalysts for CH₃OH synthesis from catalytic conversion of CO₂ using alternative feed streams with low CO₂ concentration.

ASSOCIATED CONTENT

Supporting Information

This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interests.

ACKNOWLEDGMENT

This work was supported by the U.S. Department of Energy, Office of Science, BES-Division of Materials Science and Engineering and BES-Scientific User Facilities under Contract DE-AC02-06CH11357. We acknowledge the computing resources operated by the Laboratory Computing Resource Center (ANL), and the ANL Center for Nanoscale Materials. We also thank Dr. Alex Martinson (ANL) for ALD coating.

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